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Paleomagnetism of Miocene Sedimentary Rocks around Matsushima Bay, Northeast Japan and Its Implication for the Time of the Rotation of Northeast Japan

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A paleomagnetic study of sedimentary rocks distributed in the Matsushima area, Northeast Japan, was conducted. Samples were collected from the Ajiri, Matsushima and Otsuka Formations of the early to middle Miocene Matsushimawan Group. The stability of remanence was investigated by the progressive thermal demagnetization method. The magnetization of the Matsushima and Otsuka Formations was grouped into two antipodal directions close to north-south. Its magnetic reversal sequence is correlative with that of the standard time-scale of around 16 Ma (from the Polarity Chron 5C to 5B of HARLAND *et al.* (1982)) on the basis of diatom biostratigraphy. It has been proposed by previous works that the counter-clockwise rotation of Northeast Japan occurred in the early to middle Miocene with the opening of the Japan Sea. I conclude that the rotation had completed before 16 Ma. On the other hand, eastward-deflected paleomagnetic directions have been reported from the strata in Southwest Japan which are precisely correlative with the Matsushima and Otsuka Formations by both biostratigraphy and magnetostratigraphy. This implies that the rotation of Northeast Japan took place earlier than that of Southwest Japan.

1. Introduction

In recent years, many paleomagnetic studies have been conducted regarding the time and manner of deformation of the Japan Arc at the time of the opening of the Japan Sea. OTOFUJI and MATSUDA (1983) showed that Southwest Japan (west of the Itoigawa-Shizuoka Tectonic Line) rotated clockwise with the opening of the Japan Sea between 28 Ma and 12 Ma. OTOFUJI *et al.* (1985a) compiled a large quantity of paleomagnetic and geochronologic data from Southwest Japan, and postulated that the rotation of about 50° occurred within less than 1 m.y. at around 15 Ma. For Northeast Japan (north of the Tanakura Tectonic Line), OTOFUJI *et al.* (1985b) concluded that the rotation occurred counter-clockwise between 21 Ma and 12 Ma, and TOSHA and HAMANO (1988) between 22 Ma and 15 Ma. Based on these paleomagnetic data, OTOFUJI *et al.* (1985c) proposed a "double-door" model for the opening mode of the Japan Sea.

Then, did the "double-door" open simultaneously? Paleomagnetic data from

the early to middle Miocene in Northeast Japan are still too scarce to answer this question with confidence. TANAKA *et al.* (1989) conducted a paleomagnetic study of the Shakotan Peninsula, western Hokkaido. They obtained a paleomagnetic direction of close to north from strata older than 15 Ma, and suggested that the time of the rotation of Northeast Japan, including western Hokkaido, is earlier than that of Southwest Japan. In contrast, FUJIWARA and SUGIYAMA (1986) suggested that the rotation had lasted up to about 11 Ma, from paleomagnetic data of the Oshima Peninsula, southern Hokkaido. Southwest Hokkaido and Northeast Japan are generally considered as one tectonic block, based on the continuity of geological structures, but there is no actual paleomagnetic evidence to support this assumption. Therefore, more Miocene paleomagnetic data from Northeast Japan is necessary to better understand the process of the opening of the Japan Sea.

In this paper, the results of a paleomagnetic study of the Matsushima area (Fig. 1) are reported. In this area, one of the typical Neogene diatom-bearing sections of

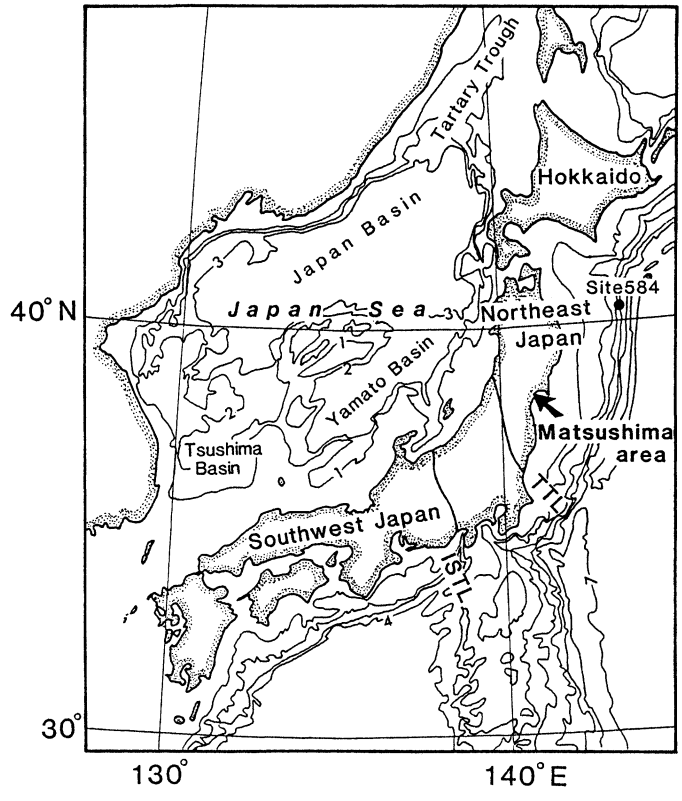
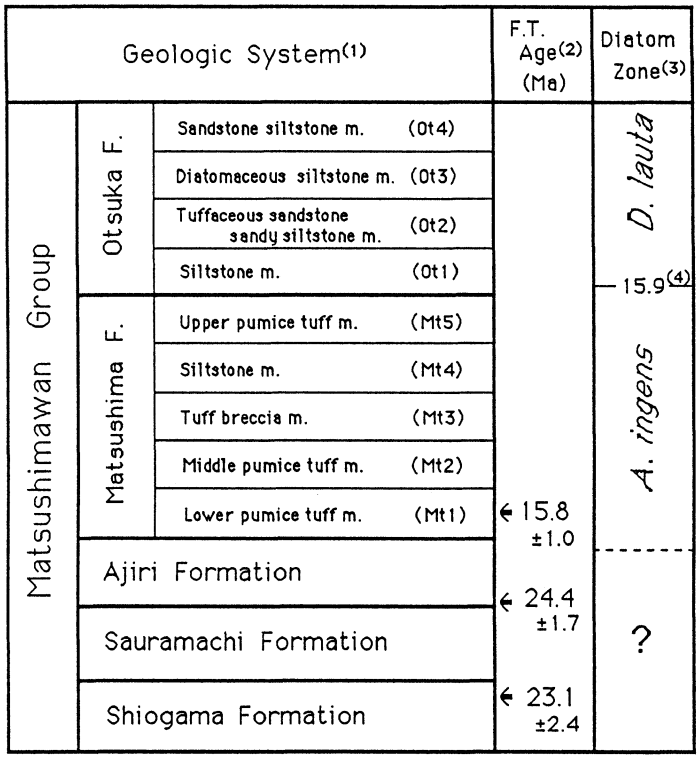


Fig. 1. Regional map of the Japan Arc and the Japan Sea. The location of the Matsushima area is shown by an arrow. Depth contours are at 1000 m intervals. Abbreviations are ISTL: the Itoigawa-Shizuoka Tectonic Line, TTL: the Tanakura Tectonic Line.

Northeast Japan is distributed. First, I have established the magnetostratigraphy of the area in order to better constrain the age of the strata with the diatom biostratigraphy of AKIBA *et al.* (1982). Next, the process of the opening of the Japan Sea is discussed based on the paleomagnetic direction and the age.

2. Geological Setting and Sampling

The Miocene Matsushimawan Group is exposed in the area around Matsushima Bay, Northeast Japan. It unconformably overlies bedrocks of pre-Tertiary age, and is unconformably covered by the Shida Group of the middle to late Miocene. The stratigraphy of the Matsushimawan Group is summarized in Fig. 2 (ISHII *et al.*, 1982, 1983a). It is divided into five lithostratigraphic units: the Shiogama, Sauramachi, Ajiri, Matsushima and Otsuka Formations, in ascending order. Data reported in this paper were obtained from the upper three formations.



(1) Ishii et al. (1982)
(2) Ishii et al. (1983b)
(3) Akiba et al. (1982)
(4) Akiba (1986)

Fig. 2. Summary of the stratigraphy of the Matsushimawan Group.

The Ajiri Formation consists mainly of alternating beds of tuff, tuffaceous sandstone and siltstone. The thickness is about 250 m. The Matsushima Formation is composed mostly of acidic pyroclastic flow deposits, which suggests that this formation was deposited in a relatively short period of time. The formation is divided lithologically into five members: lower pumice tuff (Mt1), middle pumice tuff (Mt2), tuff breccia (Mt3), siltstone (Mt4) and upper pumice tuff (Mt5). The total thickness is 350 to 400 m. The Otsuka Formation is represented mainly by thick marine siltstone intercalated with tuffaceous sandstone. The formation is lithologically divided into four members, namely siltstone (Ot1), tuffaceous sandstone sandy siltstone (Ot2), diatomaceous siltstone (Ot3) and siltstone sandstone (Ot4) members, from the bottom upward. The total thickness is about 300 m. The siltstones of the Ajiri, Matsushima and Otsuka Formation resemble each other, suggesting a similar depositional environment.

The Matsushimawan Group suffered neither vigorous tectonic disturbance nor severe alteration. The strata of the Matsushima and Otsuka Formations generally dip gently less than 10°. The beds of the Ajiri Formation monoclinally dip 20° to 30° east-southeast.

Based on the diatom biostratigraphy (AKIBA *et al.*, 1982), the strata from the uppermost part of the Ajiri Formation to the lowermost part of the Otsuka Formation are assigned to the *Actinocyclus ingens* Zone. (The Matsushima Formation and the lowermost part of the Otsuka Formation belong to the *Denticulopsis praelauta* Zone, which has been introduced recently in the upper part of the *A. ingens* Zone; Y. Yanagisawa, personal communication.) The rest of the Otsuka Formation is correlated to the *Denticulopsis lauta* Zone. The boundary exists in the lowest member (Ot1) of the Otsuka Formation. According to AKIBA (1986), the age of this boundary is 15.9 Ma. The Ajiri Formation does not yield diatoms except in the uppermost part, but contains characteristic fossil shells and plants (the Kadonosawa Fauna and the Daijima Flora) of late Early Miocene (ISHII *et al.*, 1982, 1983a). A pumice tuff from the lowest member of the Matsushima Formation (Mt1) was dated as 15.8±1.0 Ma, and that from the lowest horizon of the Ajiri Formation as 24.4±1.7 Ma by the fission track method (ISHII *et al.*, 1983b).

Samples for paleomagnetic measurements were collected by hand sampling along the stratigraphic columnar sections of ISHII *et al.* (1982, 1983a). The positions of the sampling sites on the columnar sections are shown in Fig. 3 for the Matsushima and Otsuka Formations. Relative age among sites which belong to the same geologic member but are located on different columnar sections (for example, Sites 1 and 17) is not known. For the Ajiri Formation, Sites 31, 30, 29, 32, 25, 26, 22 and 24 were located along the columnar section No. 2 of ISHII *et al.* (1983a, pp. 15–16) in ascending order. Two other sites (Sites 28 and 43) were at the uppermost part of the formation.

Samples were oriented by a magnetic compass. The number of independently oriented blocks was 3 for the sites of magnetostratigraphic studies, and 5 to 10 for the sites of tectonic consideration. Tuff and tuffaceous siltstone were collected from

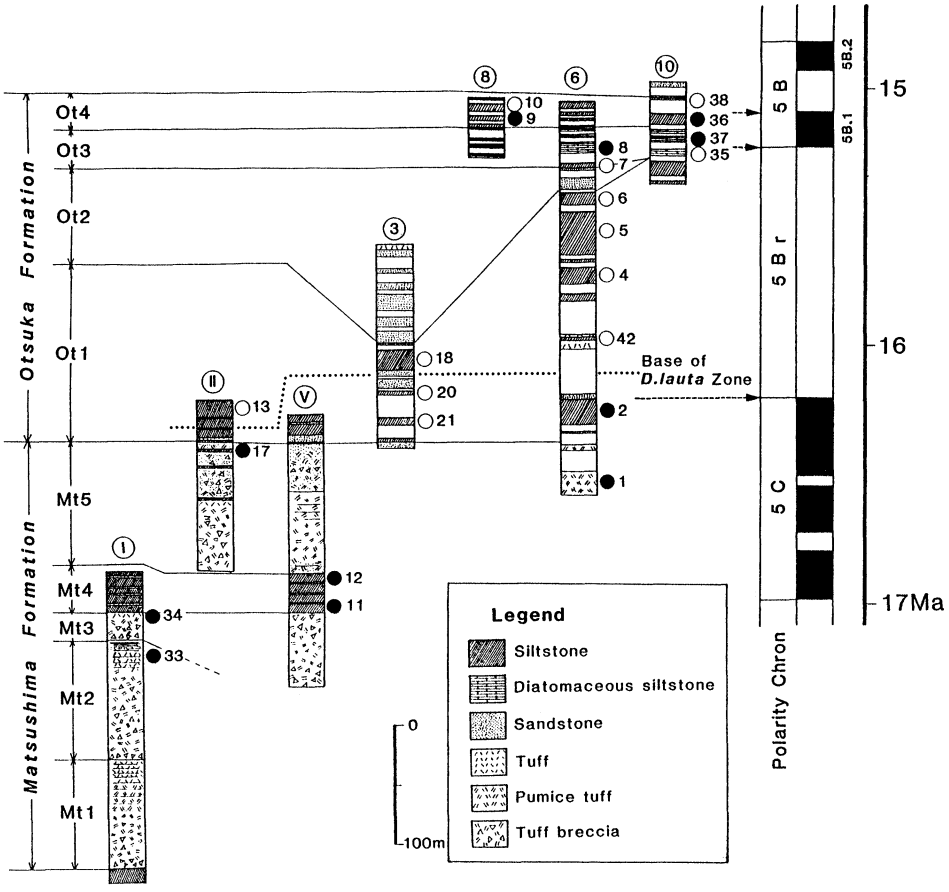


Fig. 3. Summary of the magnetostratigraphy of the Matsushima and Otsuka Formations. Solid and open circles denote normal and reversed polarity, respectively. Numbers attached to the symbols represent the names of the sampling sites. Stratigraphic columnar sections are simplified from ISHII *et al.* (1982). Numbers on the top of the columnar sections correspond to those of ISHII *et al.* (1982). The boundary of the two diatom zones, the *D. lauta* Zone and the *A. ingens* Zone, is from AKIBA *et al.* (1982). The polarity reversal time scale of HARLAND *et al.* (1982) was adopted.

the Ajiri Formation. Pumice tuff and tuff breccia of the Matsushima Formation were considered to be unsuitable for paleomagnetic studies, and we took samples from intercalated thin fine-grained layers without pumice. Samples from the Otsuka Formation consist mainly of siltstone and tuffaceous siltstone. Each sample was cut into several 2 cm cubes in the laboratory for magnetic measurements.

3. Paleomagnetic Measurements

Remanent magnetization was measured by an ScT C-113 cryogenic magnetometer and a Natsuhara-Giken SMM-86 spinner magnetometer. To investigate the stability of the remanence, three to four specimens were selected from each site and subjected to progressive thermal demagnetization. Specimens were heated in air. We also conducted stepwise alternating field demagnetization to test the stability of the remanence. However, the apparatus used could produce a maximum field of only 25 mT; in many cases, the field was too weak to remove secondary components sufficiently. The thermal demagnetization is, however, generally more successful in removing overprints than the alternating field method (e.g., ITOH, 1988). Therefore, only the results of the thermal demagnetization are described and discussed in this paper.

Two types of behavior on progressive thermal demagnetization were commonly observed on vector demagnetization diagrams. The first type (Type A) shows the stable nature of the remanence (Fig. 4). The progressive change of the vector endpoints shows a nearly linear trend. Low blocking temperature components can be removed by the first few steps. The rest of the specimens from these sites were demagnetized at an appropriate temperature between 150 and 300°C.

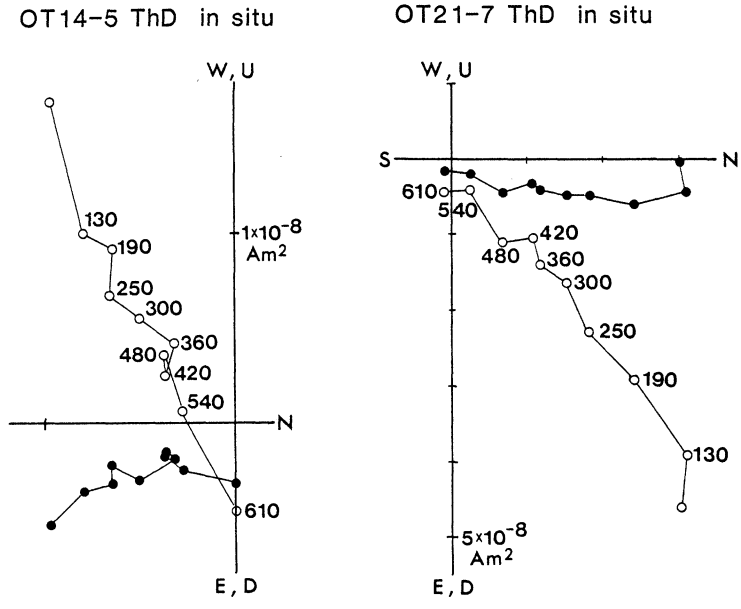


Fig. 4. Orthogonal plots of progressive thermal demagnetization of the samples which have stable magnetization (Type A). The examples are at Site 7 (left) and Site 9 (right) in the Otsuka Formation. Solid and open circles represent projections of vector endpoints on the horizontal and vertical plane, respectively, in in-situ coordinates.

On the other hand, the second type (Type B) has been affected considerably by magnetic overprinting (Fig. 5(a)). Before demagnetization, the large secondary component of present geomagnetic field direction surpassed the original, reversed component. This secondary component, which is likely of viscous origin, was reduced by demagnetization at about 200 to 300°C. In many cases, however, unstable magnetization was acquired after demagnetization at about 400°C. The origin of this unstable magnetization is not clear, but it is probably due to the growth of a new magnetic phase during the thermal demagnetization, which may acquire viscous remanent magnetization (VRM) instantaneously in the geomagnetic field in our laboratory. Only a few demagnetization steps around 300°C seem to have a trend towards the origin. Even at the sites indicating this type of thermal demagnetization behavior, fairly good convergence in direction was obtained after

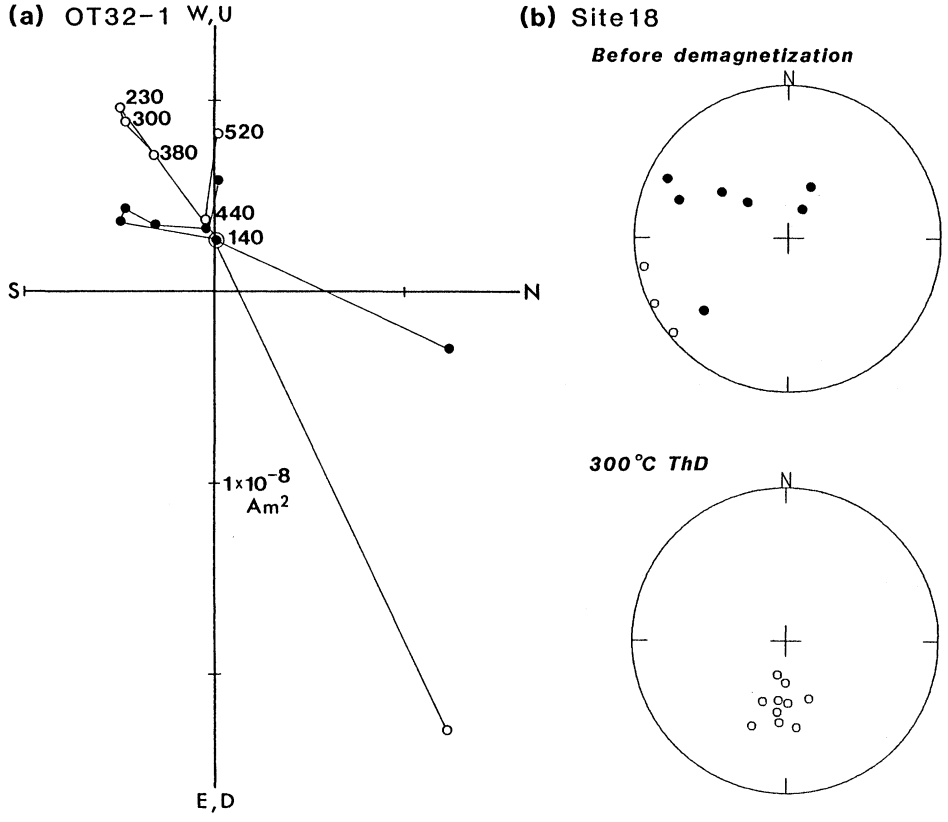


Fig. 5. (a) Thermal demagnetization behavior of a sample which suffered large magnetic overprints (Type B). The example is a specimen at Site 18, siltstone from the Otsuka Formation. Plots and symbols are the same as in Fig. 4. (b) Equal-area projection of remanent directions of samples at Site 18 before demagnetization (top) and after thermal demagnetization at 300°C (bottom). Solid and open circles denote lower and upper hemisphere directions, respectively.

demagnetization of the rest of the specimens at a proper temperature of about 300° C (Fig. 5(b)), from which site-mean directions were calculated.

A few sites which show normal polarity but do not have a clear trend towards the origin on the vector-demagnetization diagram were rejected as unstable magnetization. All of the surviving normal-polarity sites showed Type A behavior, and a few sites of reversed polarity in the Otsuka Formation. The majority of the reversed polarity sites are classified as Type B.

The magnetic direction which coincides with the present geomagnetic field direction before tilt correction is often rejected, because it might be an overprint of the present field. In this paper, however, this rejection criterion was not adopted. Through this rejection criterion, we cannot obtain the paleomagnetic direction of the normal polarity from horizontally layered strata such as the Matsushima and Otsuka Formations unless they have suffered rotation, and consequently, we cannot perform magnetostratigraphic study.

4. Results

The site-mean directions obtained from the Matsushima area are listed in Table 1. A radius of 95% confidence circle was calculated only for the sites which have five or more independently oriented samples, and the site-mean directions with the 95% confidence circles of these sites are shown in Fig. 6. Results of the paleomagnetic stratigraphy of the Matsushima and Otsuka Formations are summarized in Fig. 3.

4.1 The Otsuka and Matsushima Formations

The magnetization of the Otsuka Formation shows both normal and reversed polarity of nearly antipodal direction, which strongly indicates that they are of primary origin. The polarity reversal sequence is concordant among columnar sections. The upper part of member Ot3 and the lower part of member Ot4 of the Otsuka Formation belong to a normal polarity period. The beds of the lowermost part of member Ot1 were also deposited during a normal polarity period. The rest of the Otsuka Formation show reversed polarity. It is unrealistic to consider that the normal polarities were caused by remagnetization, because it would have been difficult to remagnetize only a part of each of the lithostratigraphic members, Ot1, Ot3 and Ot4, of the Otsuka Formation.

All six sites in the Matsushima Formation show normal polarity. The single polarity is probably due to the fact that the Matsushima Formation was deposited during a geologically short period of time as estimated from its lithology, mostly pyroclastic flow deposits. The Matsushima Formation grades into the Otsuka Formation lithologically, and there is no time gap between the two formations (ISHII *et al.*, 1982). The Matsushima Formation and the lowermost Otsuka Formation are therefore considered to belong to the same normal polarity period.

It is difficult to explain the single polarity of the Matsushima Formation by remagnetization, although not impossible. No evidence of a geologic event which would have caused remagnetization exists at the boundary of the Matsushima and

Table 1. Site-mean magnetic directions from the Matsushima area.

Site	Dec.	Inc.	P	N	α_{95}	Lat.	Lon.	Demag.
Otsuka Formation								
Ot4								
9	−17.4	53.8	N	3		73.3	83.5	200°C
10	−158.2	−41.7	R	3		70.1	−123.2	150°C
36	14.2	50.1	N	3		76.3	−101.9	240°C
38	−142.3	−36.2	R	3		52.9	−110.0	240°C
Ot3								
7	−170.3	−18.1	R	9	23.4	60.0	−58.5	300°C
8	3.6	39.8	N	3		84.6	−74.7	200°C
35	154.9	21.6	R	3		35.5	−8.2	240°C
37	−32.9	42.5	N	3		59.1	34.9	300°C
Ot1								
2	25.8	59.8	N	3		84.6	−74.7	200°C
4	−143.9	−45.4	R	3		57.9	−135.3	240°C
5	−166.5	−56.0	R	5	15.7	79.2	−125.3	300°C
6	−179.9	−58.5	R	3		77.8	−143.3	380°C
13	178.6	−47.0	R	9	14.6	80.1	−31.7	300°C
18	−175.2	−55.1	R	10	7.7	85.5	−98.5	300°C
20	118.7	−71.4	R	3		38.0	104.6	150°C
21	−134.2	−69.8	R	4		47.2	−178.3	150°C
42	−146.2	−48.4	R	2		60.8	−122.1	240°C
Matsushima Formation								
1	−12.2	39.2	N	3		77.9	32.0	200°C
11	−1.7	47.2	N	10	6.6	80.3	−30.0	200°C
12	4.8	56.5	N	3		79.2	163.8	200°C
17	4.7	50.5	N	9	11.9	82.2	−69.8	200°C
33	13.4	43.8	N	3		77.9	−123.4	200°C
34	−1.6	48.3	N	3		87.3	111.6	200°C
Ajiri formation								
22	23.1	53.6	N	10	5.3	71.0	−124.3	200°C
24	14.8	28.6	N	3		71.9	−89.4	150°C
25	36.7	60.5	N	3		56.7	−162.4	150°C
26	40.9	52.3	N	9	5.0	56.6	−132.8	200°C
28	8.9	49.5	N	3		81.5	−132.5	150°C
29	33.6	60.3	N	10	4.7	64.1	−146.7	200°C
30	23.8	51.1	N	3		68.8	−145.7	150°C
31	15.1	56.7	N	3		73.6	−169.7	200°C
32	15.2	61.8	N	3		69.8	176.7	150°C
43	12.7	32.1	N	5	13.2	66.6	−70.8	200°C

Dec. and Inc. are mean declination and inclination after tilt correction; P is the magnetic polarity (N: normal, R: reversed); N is the number of independently oriented samples; α_{95} is a radius of 95% confidence circle in degrees (listed only for the sites which have five or more independently oriented samples); Lat. and Lon. are latitude (+north) and longitude (+east) of the VGP (virtual geomagnetic pole) position (north-seeking pole). Demag. is the temperature of thermal demagnetization performed.

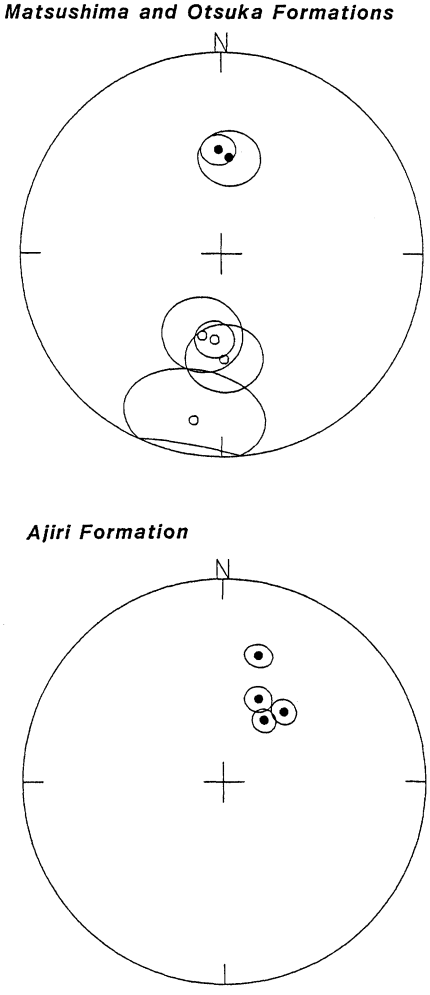


Fig. 6. Site-mean directions of (top) the Matsushima and Otsuka Formations and (bottom) the Ajiri Formation. Only the sites which have five or more independently oriented samples are plotted. Plots and symbols are the same as in Fig. 5(b).

Otsuka Formations or at the polarity boundary in the lowermost Otsuka Formation. Remagnetization could have occurred after the deposition of the Otsuka Formation, but it would have been difficult to remagnetize only the Matsushima Formation.

Viscous remanence acquired over the Brunhes Epoch also cannot explain the single polarity. All samples from the Matsushima Formation showed a linear trend up to 500°C or higher on the vector demagnetization diagram (Type A), but such a

high VRM unblocking temperature has not been known. Thermal demagnetization of about 300° C can remove VRM (YAMAZAKI, 1986; KENT and MILLER, 1987). The theoretical consideration of PULLAIAH *et al.* (1975) predicts that the VRM acquired for about 1 m.y. at 20° C has an unblocking temperature of about 250° C (magnetite) or about 350° C (hematite). If the VRM is acquired at higher temperature, the unblocking temperature increases, but the rocks of the Matsushima Formation contain no evidence of elevated temperature after their deposition, such as low-grade metamorphism.

The formation-mean direction, a declination of 5.0° and inclination of 45.9° ($\alpha_{95}=12.2°$), was derived from the site-mean directions of the six sites in the Matsushima and Otsuka Formations which have five or more independently oriented samples. The direction was almost north-south, or slightly eastward-deflected although the 95% confidence circle includes the axial dipole field direction. The declinations of the sites for magnetostratigraphic purposes also have a similar tendency (Table 1).

4.2 The Ajiri Formation

All sites of the Ajiri Formation have stable remanence of normal polarity (Type A). The formation-mean direction, $D=25.2°$, $I=50.2°$ and $\alpha_{95}=16.9°$, was obtained from the site-mean directions of the four sites which have five or more independently oriented samples. The direction seems to be deflected east slightly although the 95% confidence circle includes the axial dipole field direction. A similar trend can be recognized at the sites for magnetostratigraphy.

An explanation for the single polarity is that the Ajiri Formation had been deposited within a normal polarity period. The Ajiri Formation is estimated to have had fairly high sedimentation rate from its lithology, alternating beds of tuff, sandstone and siltstone. The single polarity cannot be caused by VRM as in the case of the Matsushima Formation.

However, a suspicion of remagnetization remains. The magnetic directions before tilt correction are close to the direction of the axial dipole field, and they become eastward-deflected after the correction. As the strata of the Ajiri Formation are monoclinal, the fold test cannot be conducted. There would be a slight tectonic gap between the Ajiri Formation and the Matsushima Formation. The strata of the Ajiri Formation dip 20° to 30°, and the dip of the beds of the Matsushima Formation is less than 10° in general. Remagnetization might have occurred in this gap. The time gap between these formations, however, would be very short because they belong to the same diatom zone.

One way to solve the above problem is to investigate whether the strata of other areas which are correlated with the Ajiri Formation also have the same magnetic direction or not. If not, the possibility of remagnetization increases. Unfortunately, this test has not yet succeeded. The Tsukinoki Formation, which is correlated with the Ajiri Formation by fossil evidence, is distributed south of Sendai City, about 30 km southwest of the present study area (OIDE and FUJITA, 1975; KITAMURA *et al.*, 1986). Tuffs and siltstones collected at five sites were submitted for paleomagnetic

study, but progressive thermal demagnetization could not separate a primary remanence from intensive multicomponent secondary magnetization (Yamazaki, unpublished data).

At present, we cannot safely say whether the remanence of the Ajiri Formation is stable secondary magnetization, or whether the direction of the remanence before tilt correction is accidentally close to that of the axial dipole field. Therefore, I exclude the Ajiri Formation from the following discussion in this paper.

5. Discussion

5.1 Magnetostratigraphy

Magnetostratigraphy provides fundamental data for precise chronology and world-wide correlation of strata. In this section, I will attempt to correlate the polarity reversal sequence of the Matsushima and Otsuka Formations with that of the standard time-scale of HARLAND *et al.* (1982).

The Miocene diatom datum levels in the temperate to subarctic northwest Pacific have not yet been tied directly to the paleomagnetic stratigraphy. But, the top of the *D. lauta* Zone (the base of the *D. hyalina* Zone) has been placed in the Polarity Chron 5B through correlation with the low-latitude diatom datum level (the last occurrence of *Annellus californicus*) in California (BARRON, 1981; BARRON and KELLER, 1983) which has been tied directly to the paleomagnetic stratigraphy in the equatorial Pacific (OPDYKE *et al.*, 1974; BURCKLE, 1978). The upper part of the *A. ingens* Zone is correlative with the Polarity Chron 5C in the same manner (BARRON, 1985). Therefore, the reversed polarity period of the majority of the Otsuka Formation is correlative with the Polarity Chron 5Br (15.23–16.20 Ma) (Fig. 3).

The Matsushima Formation was deposited during the Polarity Chron 5C (16.20–16.98 Ma). The normal polarity at the upper part of member Ot3 and the lower part of member Ot4 probably corresponds to the Polarity Chron 5B.2 (15.09–15.23 Ma). There is a possibility that the normal polarity corresponds to Chron 5B.1. In this case, Chron 5B.2 would exist in member Ot2, from which no sample was taken. But I feel that the former correlation is better, because member Ot2, which consists mainly of sandstone, would have accumulated faster than other members, which consist mainly of siltstone.

The boundary of the two diatom zones, the *D. lauta* Zone and the *A. ingens* Zone, is on or just above the boundary of the Polarity Chrons 5C and 5Br. In the Northwest Pacific, this is the first direct correlation of that diatom datum level with the magnetostratigraphy.

An effort to correlate Neogene diatom biostratigraphy with magnetostratigraphy in the Northwest Pacific was made at Site 584 of the Deep Sea Drilling Project off Northeast Japan (NIITSUMA, 1986; AKIBA, 1986). The age of the bottom of the core was inferred to be about 16 Ma from diatom biostratigraphy, but the oldest magnetic polarity chron correlated with the standard time-scale was the Chron 5AA (12.76–12.94 Ma). A low sedimentation rate with possible hiatuses and

intensive remagnetization prevented the establishment of the magnetostratigraphy below it.

Based on the above correlation, we can estimate the mean sedimentation rate of the Otsuka Formation. The rate of 25 cm/1000 yr is derived from the thickness of the Otsuka Formation, about 300 m (ISHII *et al.*, 1982), and the elapsed time of about 1.2 m.y. This rate is of the same order of magnitude as the sedimentation rate in the forearc basins around Japan (OCEANOGRAPHICAL SOCIETY OF JAPAN, 1985; IKEHARA, 1988).

5.2 Implication for the opening of the Japan Sea

This study revealed that the Matsushima and Otsuka Formations, which were dated as about 16 Ma by magnetostratigraphy and biostratigraphy, have paleomagnetic directions of close to north-south. This implies that the counter-clockwise rotation of Northeast Japan with the opening of the Japan Sea had completed before 16 Ma. An undeflected paleomagnetic direction of about the same age was obtained at Shakotan Peninsula, western Hokkaido (TANAKA *et al.*, 1989). This suggests that Northeast Japan and western Hokkaido belong to the same tectonic block.

In Southwest Japan, several paleomagnetic studies of strata which are precisely correlative with the Matsushima and Otsuka Formations by both biostratigraphy and magnetostratigraphy have been reported.

HAYASHIDA and ITO (1984) obtained largely eastward-deflected declination, about 50°, from the upper Ichishi Group, which corresponds magnetostratigraphically to the Chrons from 5Cr to 5Br (17.6–15.2 Ma) and biostratigraphically to the Zone N8 of BLOW (1969) (15.5–16.5 Ma; SAITO, 1984). They concluded that the majority of the clockwise rotation of Southwest Japan occurred between 16 Ma and 12 Ma. HAYASHIDA (1986) studied other two groups of the Setouchi Miocene Series and reached the same conclusion. At 16 Ma, the major drift of Southwest Japan had not yet occurred. This implies that the rotation of Northeast Japan took place earlier than that of Southwest Japan.

ITOH (1986) showed that the Yatsuo Group in the easternmost part of Southwest Japan has a little (about 13°) eastward-deflected paleomagnetic direction. ITOH (1988) concluded that a differential rotation between the eastern part and the main block of Southwest Japan occurred between 15 and 12 Ma with the opening of the Japan Sea. Diatom biostratigraphy showed that the upper part of the Yatsuo Group (the Higashibessho Formation) belongs to the *D. lauta* Zone (ITOH, 1988; ITOH and HAYAKAWA, 1988). Thus, the Higashibessho Formation can be correlated directly with the Otsuka Formation by diatom biostratigraphy. The eastward-deflected direction was obtained from the former, but not from the latter. This strongly indicates that the rotation of Northeast Japan was completed earlier than that of Southwest Japan.

The Japan Sea has four major sedimentary basins, namely the Japan Basin, the Yamato Basin, the Tartary Trough and the Tsushima Basin (TAMAKI, 1986). As briefly summarized by KONO (1987), the age of the basement of the Japan Sea

remains controversial. The opening of the northeastern half of the Japan Basin and the Tartary Trough would have caused the counterclockwise rotation of Northeast Japan, and the opening of the southwestern half of the Japan Basin, the Yamato Basin and the Tsushima Basin would have raised the clockwise rotation of Southwest Japan. Our paleomagnetic data suggest that the former basins opened earlier than the latter.

Although the 95% confidence ellipse includes the axial dipole field direction, the declinations from the Matsushima area seem to be slightly deflected east. The declinations from the strata of 12 Ma or younger at Oga Peninsula, Northeast Japan (TOSHA and HAMANO, 1988) and the strata of 15 Ma or younger at Shakotan Peninsula, western Hokkaido (TANAKA *et al.*, 1989) have the same tendency. There is a possibility that Northeast Japan was slightly rotated clockwise after its counter-clockwise rotation with the opening of the Japan Sea. However, much more paleomagnetic data are needed to confirm this.

6. Conclusion

The paleomagnetic study of the Matsushima area, Northeast Japan, leads to the following conclusions.

1) The magnetic polarity reversal sequence of the Matsushima and Otsuka Formations is correlative with that of around 16 Ma of the standard time-scale (from the Polarity Chron 5C to 5B of HARLAND *et al.*, 1982). The boundary of the two diatom zones, the *D. lauta* Zone and the *A. ingens* Zone, is on or just above the boundary of the Polarity Chrons 5C and 5Br (16.2 Ma).

2) The paleomagnetic directions of the Matsushima and Otsuka Formations are close to north-south. This implies that the counter-clockwise rotation of Northeast Japan, which is known to have occurred with the opening of the Japan Sea in early to middle Miocene time, had completed before 16 Ma.

3) The rotation of Northeast Japan had occurred earlier than that of Southwest Japan. An eastward-deflected paleomagnetic direction has been reported from strata in Southwest Japan which are correlated precisely with the Matsushima and Otuska Formations by both biostratigraphy and magnetostratigraphy.

4) The Ajiri Formation, which is overlain by the Matsushima Formation, also showed a north or slightly eastward-deflected magnetic direction. However, the suspicion that the magnetization of the Ajiri Formation is of secondary origin still remains.

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